SAGINAW VALLEY SEEDBANK

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Organization Overview:

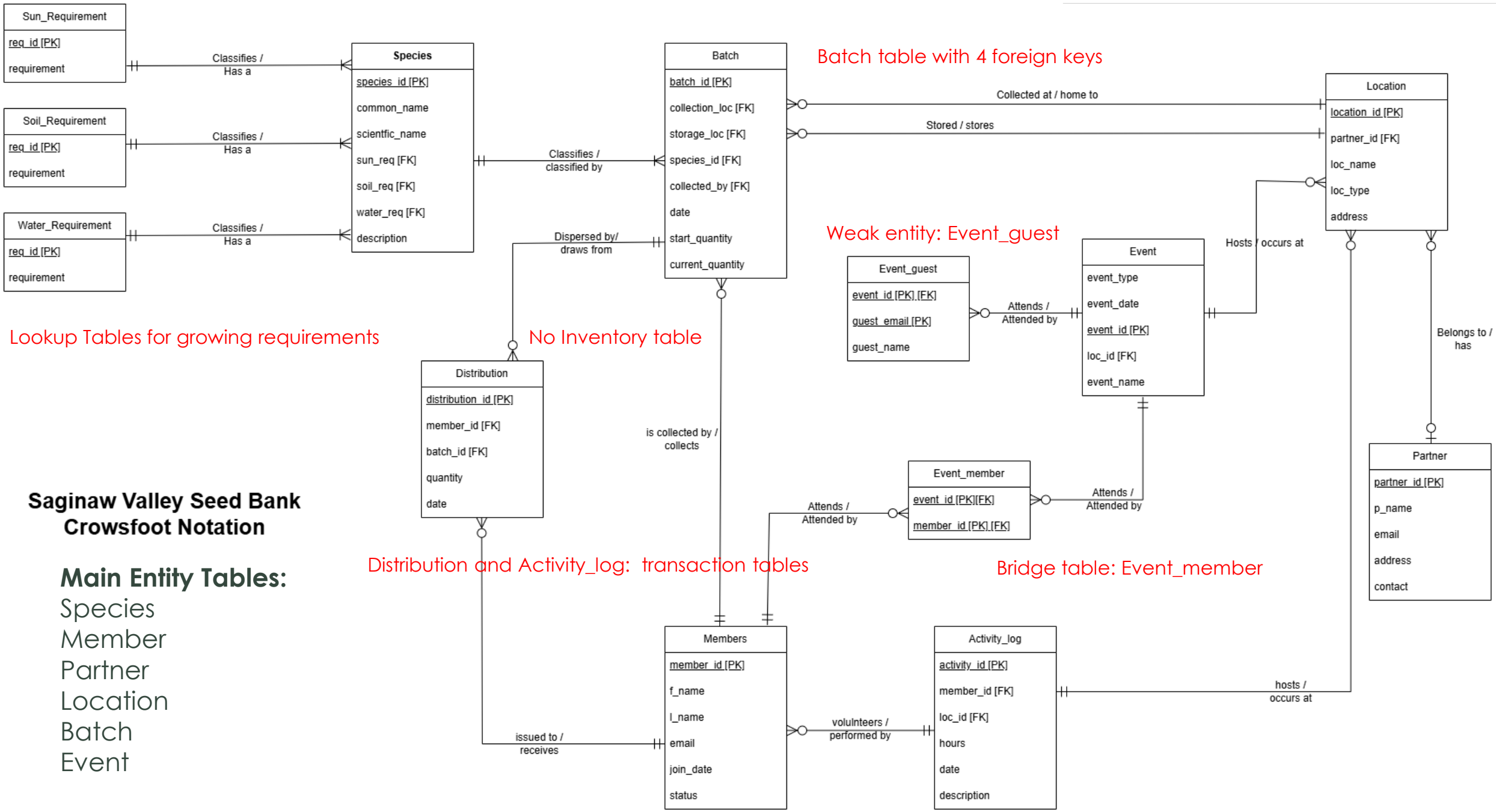
Type of Work

- Seedbank
- Knowledge Repository
- Member Tracking
- Community Education
- Partner Relations

Organization Overview:

Database Requirements

- Species database Module
- Seed inventory module
- Seed batch module
- Member / Volunteer module
- Distribution module
- Partner module
- Education module
- Reporting / Analytics



Saginaw Valley Seed Bank
Crowsfoot Notation

Main Entity Tables:

Species
Member
Partner
Location
Batch
Event

SQL: Sample Data

Cut tables

Event_attendence

event_id	member_id	name	email
1	1	Sarah Kowalski	sarah.k@email.com
1	2	Marcus Trent	mtrent@email.com
1	4	James Obi	jobi@email.com
1	5	Priya Nair	pnair@email.com
1		Karen Walker	walkerin@gmail.com
2	1	Sarah Kowalski	sarah.k@email.com
2	6	Tom Walcott	twalcott@email.com
2	8	Raj Patel	rpatel@email.com
3	2	Marcus Trent	mtrent@email.com
3	3	Dana Flores	dflores@email.com

Guest

event_id	email	name

Inventory

batch_id	quantity	storage_location

Species before…

Species

species_id	scientific_name	common_name	sun_req	soil_req	water_req
1	Asclepias tuberosa	Butterfly Weed	Full Sun	Sandy/Dry	Low
2	Echinacea purpurea	Purple Coneflower	Full Sun/Part Shade	Loamy	Medium
3	Lobelia cardinalis	Cardinal Flower	Part Shade	Moist/Rich	High
4	Andropogon gerardii	Big Bluestem	Full Sun	Sandy/Clay	Low-Medium
5	Baptisia australis	Blue Wild Indigo	Full Sun	Well-drained	Low
6	Monarda fistulosa	Wild Bergamot	Full Sun	Loamy	Medium
7	Sorghastrum nutans	Indiangrass	Full Sun	Sandy/Clay	Low
8	Penstemon digitalis	Foxglove Beardtongue	Full Sun/Part Shade	Loamy	Medium
9	Rudbeckia hirta	Black-eyed Susan	Full Sun	Well-drained	Low-Medium
10	Zizia aurea	Golden Alexanders	Part Shade	Moist	Medium-High

…and after

species_id	common_name	scietific_name	sun_req	soil_req	water_req	description
1	Butterfly Weed	Asclepias tuberosa	1	1	1	Full sun, dry soil specialist; monarch host plant
2	Purple Coneflower	Echinacea purpurea	2	2	3	Adaptable prairie perennial; medicinal uses
3	Cardinal Flower	Lobelia cardinalis	3	5	5	Shade-tolerant; brilliant red; hummingbird magnet
4	Big Bluestem	Andropogon gerardii	1	3	2	Dominant tallgrass prairie grass; deep roots
5	Blue Wild Indigo	Baptisia australis	1	4	1	Drought tolerant legume; slow to establish
6	Wild Bergamot	Monarda fistulosa	1	2	3	Fragrant lavender blooms; excellent for bees
7	Indiangrass	Sorghastrum nutans	1	3	1	Warm season grass; golden fall color

Event_guest

event_id	guest_email	guest_name
1	dsharma@gmail.com	Dev Sharma
1	walkerin@gmail.com	Karen Walker
2	lgravs@hotmail.com	Lila Graves
3	ohussain@email.com	Omar Hussain
3	tbrooks@yahoo.com	Tanya Brooks
4	mjanssen@gmail.com	Mike Janssen
5	rdelgado@email.com	Rosa Delgado
7	fcalloway@yahoo.com	Finn Calloway
8	ytanaka@gmail.com	Yuki Tanaka
9	bcromwell@email.com	Beth Cromwell

distribution

dist_id	batch_id	member_id	quantity	date
1	1	1	10	2024-03-20
2	1	3	5	2024-04-02
3	2	2	15	2024-04-15
4	3	1	10	2024-07-01
5	4	6	5	2024-08-05
6	5	7	20	2024-08-10
7	6	10	10	2024-07-20
8	8	2	15	2024-05-15
9	9	1	10	2024-06-10
10	3	4	5	2024-08-22

batch

batch_id	collection_loc	storage_loc	species_id	collected_by	date	start_quantity	current_quantity
1	1	4	1	1	2024-03-10	50	50
2	3	4	3	6	2024-04-02	100	100
3	2	4	6	2	2024-04-15	75	75
4	1	5	11	1	2024-07-01	30	30
5	6	4	6	7	2024-08-05	200	200
6	7	4	4	6	2024-03-25	150	150
7	10	4	20	10	2024-07-20	10	10
8	2	4	17	2	2024-05-10	80	80
9	1	5	2	1	2024-06-03	70	70
10	4	4	9	NULL	2024-08-22	25	25

SQL Queries pt 1

```
SELECT l_name AS "Last Name", f_name AS "First Name", COUNT(batch_id) as "Count"
FROM member, batch
WHERE collected_by = member_id
GROUP BY l_name, f_name
HAVING COUNT(batch_id) > ( SELECT AVG(BATCH_COUNT)
                    FROM (      SELECT COUNT(batch_id) AS BATCH_COUNT
                                 FROM batch
                                GROUP BY collected_by) AS CB
                    )
```

Last Name	First Name	Count
Kowalski	Sarah	3
Trent	Marcus	2
Walcott	Tom	2

```
SELECT common_name AS "Species", SUM(b.start_quantity) AS "Seeds Collected", SUM(d.quantity) AS "Seeds Distributed"
FROM species s
JOIN batch b ON s.species_id = b.species_id
LEFT JOIN distribution d ON b.batch_id = d.batch_id
GROUP BY common_name;
```

Species	Seeds Collected	Seeds Distributed
Butterfly Weed	100	15
Cardinal Flower	100	15
Wild Bergamot	350	35
White Trillium	30	5
Big Bluestem	150	10
Mayapple	10	NULL
Canada Goldenrod	80	15
Purple Coneflower	70	10
Black-eyed Susan	25	NULL

SQL Queries pt 2

```
SELECT s.common_name, sun.requirement AS sun_req, water.requirement AS water_req
FROM Species s, Sun_Requirement sun, Water_Requirement water
WHERE s.sun_req = sun.req_id
  AND s.water_req = water.req_id
  AND sun.requirement = 'Full Sun'
  AND water.requirement = 'Low'
```

common_name	sun_req	water_req
Butterfly Weed	Full Sun	Low
Blue Wild Indigo	Full Sun	Low
Indiangrass	Full Sun	Low
Little Bluestem	Full Sun	Low

```
SELECT event_name, COUNT(DISTINCT m.member_id) + COUNT(DISTINCT g.guest_email) AS 'Total Attendees'
FROM Event e
LEFT JOIN Event_member m ON e.event_id = m.event_id
LEFT JOIN Event_guest g ON e.event_id = g.event_id
GROUP BY event_name
ORDER BY 'Total Attendees' DESC
LIMIT 3
```

ERROR – SHOULD BE `` nor '' on Total Attendees

event_name	Total Attendees 1
Spring Seed Swap	5
Native Plant Workshop	4
Fall Harvest & Processing	3
Spring Cleaning	0

event_name	Total Attendees 1
	5
	4
ng	3
	3
off	3
	2
nting	2
School Outreach Demo	2

Problems with SQL

Rigid Schema

Some plants fit nicely into the sun, soil, and water requirements, but others require more information or different fields. How can we adjust the species table to be more flexible when it comes to growing attributes?

Joins are confusing and slow

Batch requires 4 foreign keys, making it difficult queries complicated for volunteers to write. How can we simplify the database to make querying easier?

Nesting / Hierarchies are awkward

Attendance tracking requires two separate tables. How can we track event attendance and walkins/ guests without multiple tables and complicated joins?

MongoDB:
Updated Tables

Events

- Nesting removed need for 2 separate attendance tables


```
_id: ObjectId('69e7e5a12ed556d1a0122423')
event_id : 1
event_name : "Spring Seed Swap"
event_date : "2024-04-06"
event_type : "member"
location : "Dow Diamond Community Garden"
member_attendance : Array (3)
  0: "Sarah Kowalski"
  1: "Marcus Trent"
  2: "James Obi"
guest_attendance : Array (2)
  0: Object
    name : "Dev Sharma"
    email : "dsharma@gmail.com"
  1: Object
    name : "Karen Walker"
    email : "walkerin@gmail.com"
```


How can we track event attendance and walkins/ guests without multiple tables and complicated joins?

Species

- Nesting 'requirements' allowed the deletion of the three lookup tables
- Other attributes, like drought_tolerance, can be added to relevant documents.


```
_id: ObjectId('69e7e9da707fb35e0f90398b')
species_id : 1
common_name : "Butterfly Weed"
scientific_name : "Asclepias tuberosa"
requirements : Object
  sun : "Full Sun"
  soil : "Sandy / Dry"
  water : "Low"
description : "Full sun, dry soil specialist; monarch host plant"
drought_tolerant : true
```

```
_id: ObjectId('69e7e9da707fb35e0f90398c')
species_id : 2
common_name : "Purple Coneflower"
scientific_name : "Echinacea purpurea"
requirements : Object
  sun : "Full Sun / Part Shade"
  soil : "Loamy"
  water : "Medium"
description : "Adaptable prairie perennial; medicinal uses"
```


How can we simplify the database to make querying easier?

Location

- The location table was removed, and locations are stored in batch, events, and partners seperatedly
- Other attributes, like drought_tolerance, can be added to relevant documents.

Batch

- Without all the reference tables, batches are much easier to read (and query... Maybe)


```
_id: ObjectId('69e7f1a8707fb35e0f9039ac')
batch_id : 1
species : "Butterfly Weed"
collection_loc : Object
  name : "Crow Island Woods Preserve"
  type : "natural area"
  address : "Crow Island Rd, Saginaw MI"
storage_loc : Object
  name : "SVSU Seedbank Storage"
  address : "7400 Bay Rd, University Center MI"
collected_by : "Sarah Kowalski"
date : "2024-03-10"
start_quantity : 50
current_quantity : 50
```


How can we simplify the database to make querying easier?

Mongo Queries

```
db["Species"].find({$and:
        [
          {"requirements.sun":"Full Sun"},
          {"requirements.water":"Low"}
        ]
      })
```

```
< {
    _id: ObjectId('69e7e9da707fb35e0f90398b'),
    species_id: 1,
    common_name: 'Butterfly Weed',
    scientific_name: 'Asclepias tuberosa',
    requirements: {
      sun: 'Full Sun',
      soil: 'Sandy / Dry',
      water: 'Low'
    },
    description: 'Full sun, dry soil specialist; monarch host plant',
    drought_tolerant: true
  }
  {
```

```
db['Events'].aggregate([
    {$group:{
      _id:"$event_name",
        attendence:{
            $sum:{
            $add:[
                {$size: "$guest_attendance"},{$size: "$member_attendance"} ]
                }
              }
            }
    },
    {$sort:{attendence:-1}},
    {$limit:3}
    ])
```

```
{
  _id: 'Spring Seed Swap',
  attendence: 5
}
{
  _id: 'Native Plant Workshop',
  attendence: 4
}
{
  _id: 'Prairie Restoration Day',
  attendence: 3
}
```

Mongo Queries 2

- db["Events"].aggregate([
- {$unwind:"$guest_attendance"},
- {$project:{_id:0,
  name:"$guest_attendance.name",
  email:"$guest_attendance.email"}}
-])

```
< {
    name: 'Yuki Tanaka',
    email: 'ytanaka@gmail.com'
  }
  {
    name: 'Rosa Delgado',
    email: 'rdelgado@email.com'
  }
  {
    name: 'Omar Hussain',
    email: 'ohussain@email.com'
  }
  {
    name: 'Tanya Brooks'
```

- db['Species'].aggregate([
- {
- $lookup:{
- from: "Batch",
- localField: "common_name",
- foreignField: "species",
- as: "species"
- }
- },
- {$match:{"species":{$size:0}}}
-
-])

```
  ])
{
  _id: ObjectId('69e7e9da707fb35e0f903994'),
  species_id: 10,
  common_name: 'Golden Alexanders',
  scientific_name: 'Zizia aurea',
  requirements: {
    sun: 'Part Shade',
    soil: 'Moist',
    water: 'Medium / High'
  },
  description: 'Early spring bloomer; supports specialist bees',
  species: []
}
{
  _id: ObjectId('69e7e9da707fb35e0f90399d'),
  species_id: 19,
  common_name: 'Little Bluestem',
  scientific_name: 'Schizachyrium scoparium',
  requirements: {
    sun: 'Full Sun',
    soil: 'Sandy / Dry',
    water: 'Low'
```

SQL vs MongoDB (NoSQL)

MongoDB

- Pros
  - Readability –no foreign keys
  - Flexibility – fields can very, schema changes are easy
  - Simplicity – less joins, no bridge tables, easier queries
- Cons
  - Lost data integrity
  - Data duplication – no referential integrity, no single source of truth
  - Lack of consistency complicates queries

SQL

- Pros
  - Data integrity – enforced through normalization / lookup tables
  - Standardization – you know exactly what you're getting from the data
  - Widely used and well supported
- Cons
  - Rigid – schema is hard to change
  - Joins can get really slow
  - Scales poorly

Conclusion

- For the seedbank, I think MongoDB is the better fit. Its more intuitive, and its document structure makes sense for data like lists, transactions, and text dense data like the species documentation.
- The lack of standardization on plant properties could be a problem from a lookup perspective, but could be managed with controls (drop down boxes for input, foe example)
- For transaction management, part tracking, accounting, etc, SQL is still probably better.